



LUND
UNIVERSITY

Numerical Analysis for E3
Numerical Analysis, Centre for Mathematical Sciences

Achim Schroll, March 2003

Homework assignment 3

Delivery date: Tuesday April 8th, 2003, no later than 18:00h.

Topics of this assignment: Linear systems of equations; Least squares; Condition number.

1. Linear Algebra. Consider the linear system $Ax = b$, with

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 3 & -5 & 0 \\ 1 & 7 & 2 & z \\ y & 10 & y & z \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 2 \\ -2 \\ 6 \\ y \end{pmatrix}$$

Determine the values of y and z such that the system $Ax = b$

- a. has a unique solution,
- b. has infinitely many solutions,
- c. has no solution.

2. Condition of the Vandermonde matrix. In MATLAB the Vandermonde matrix

$$V = \begin{pmatrix} t_0^n & t_0^{n-1} & \dots & 1 \\ t_1^n & t_1^{n-1} & \dots & 1 \\ \vdots & \vdots & & \vdots \\ t_n^n & t_n^{n-1} & \dots & 1 \end{pmatrix}$$

is generated by `vander(t)`. The goal is to investigate how the condition number of this matrix depends on its dimension.

Task 2a): For `t = linspace(-1,1,n)'` and various dimensions $1 \leq n \leq 35$ compute the condition number (in 2-norm) and illustrate the dependence on n in a semilogarithmic plot. Find a formula for this dependence. Useful MATLAB commands are `cond` and `semilogy`.

Task 2b): Now assume you are solving a linear system of equations $Vx = b$ (where V is the Vandermonde matrix) on a computer using IEEE DP arithmetic. Give an upper bound of how large relative error you have to expect when solving the system with 16 unknowns.

3. An ill-conditioned linear system. The Hilbert matrix

$$H = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \vdots & \vdots & \vdots & & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1} \end{pmatrix}$$

is obviously symmetric. A stable algorithm for solving the linear system $Hx = b$ is via Cholesky decomposition:

For every symmetric, positive definite matrix A there is a unique factorization

$$A = LDL^T,$$

where L is a lower unit triangular matrix and $D = \text{diag}(d_1, d_2, \dots, d_n)$ with $d_i > 0$ for all $i = 1, 2, \dots, n$. This factorization is called the **Cholesky decomposition**. Note that a Cholesky decomposition corresponds to an LR decomposition with $R := DL^T$, but it is cheaper to compute.

Task 3a): Suppose you have a Cholesky decomposition given by $A = LDL^T$. Describe how you would solve the linear system $Ax = b$.

In MATLAB the Cholesky decomposition is computed in a slightly different form. The command `R = chol(A)` computes $R = (LD^{1/2})^T$, such that

$$R^T R = LD^{1/2}(LD^{1/2})^T = LDL^T = A,$$

with $D^{1/2} = \text{diag}(\sqrt{d_1}, \sqrt{d_2}, \dots, \sqrt{d_n})$.

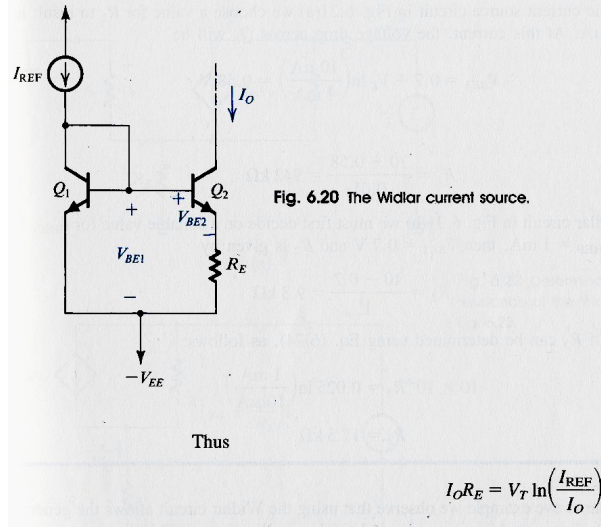
Task 3b): How can this decomposition $A = R^T R$ be used to solve the linear system $Ax = b$?

Task 3c): Solve the Hilbert system $Hx = b$ with right hand side $b = (\frac{1}{n}, \frac{1}{n+1}, \dots, \frac{1}{2n-1})^T$ and dimension $n = 12$. Apply MATLAB's Cholesky factorization `R = chol(H)`. The exact solution of the system $Hx = b$ is $x = (0, 0, \dots, 0, 1)^T$. What is the relative error of your approximation in max-norm $\|\cdot\|_\infty$? Can you explain this result?

4. Average property of least squares. Show that a straight line $y = mx + b$ which is fitted in the least-squares sense to n measurement points $(x_i; y_i)$ passes the point $(x^*; y^*)$ with

$$x^* = \sum_{i=1}^n \frac{x_i}{n} \quad \text{and} \quad y^* = \sum_{i=1}^n \frac{y_i}{n}.$$

Hint: formulate First the normal equations.



Widlar source current and its governing equation
(Taken from *Microelectronic Circuits*, Sedra & Smith, third edition).

5. Widlar source current. A Widlar current source can be described by the following equations

$$I_O R_e = V_t \ln\left(\frac{I_{ref}}{I_o}\right)$$

As V_t depends on the temperature of operation of the circuit this device can be used as a thermometer if we measure some of its parameters at a constant temperature. The relation between V_t and the temperature T in Kelvin degrees is given by

$$V_t = \frac{KT}{q}$$

where $q = 1.60218 \times 10^{-19}$ (charge of the electron) and $K = 1.38066 \times 10^{-23}$ (Boltzman constant).

Let us assume we measured the following values of I_{ref} , R_e , and I_o :

I_o (μA)	I_{ref} (mA)	R_e (K Ω)
7.859763	1	16
11.210700	2	12
11.165746	3	13
7.472788	10	25
9.294492	12	20
13.264235	17	14

Determine the temperature from these measurements.